

INFOSYS SPRINGBOARD

Documentation

Milestone-3

Name:- Soham Badgujar

Project Title:- Interactive Analytics Dashboard for Global Income Distribution

Project Statement and Outcomes:-

The Interactive Analytics Dashboard for Global Income Distribution project aims to develop an interactive Power BI dashboard that visualizes income inequality data across various countries and regions. This project involves collecting, preprocessing, and structuring global income distribution and economic data to enable effective visualization. The dashboard will provide users with insights into income disparities over time, country-wise comparisons, and patterns of inequality globally.

The outcome will be a user-friendly dashboard, embedded within a Streamlit web application, offering an intuitive interface for economists, researchers, policymakers, and the general public. This platform will help users explore global income inequality, understand disparities across different regions, and analyze trends in wealth distribution. The project will conclude with comprehensive testing to ensure dashboard accuracy and usability, supported by detailed documentation for future reference and enhancements.

1. Introduction

In the modern data-driven world, organizations rely on analytical platforms to interpret complex datasets and generate meaningful insights. The objective of this project is to develop an interactive analytics platform capable of exploring global income distribution data and generating predictive insights.

The project initially used Microsoft Power BI for visualization of global inequality indicators. However, to enhance flexibility, interactivity, and machine learning integration, the system was extended into a full web application using Streamlit.

The resulting system is called the Global Income Intelligence Platform, which combines business intelligence dashboards, machine learning models, and advanced data visualization techniques into a single unified platform.

The application also utilizes several Python libraries including Pandas, Plotly, Matplotlib, Seaborn, and machine learning algorithms from Scikit-learn.

2. Objectives of Milestone 3

The main objectives of this milestone are:

- Transform the analytics dashboard into a deployable web platform
- Integrate business intelligence tools with Python-based analytics
- Implement interactive data visualization features
- Enable machine learning predictions and forecasting
- Provide automated insights from datasets
- Create an intuitive and responsive user interface

The milestone demonstrates how a traditional data dashboard can evolve into a complete data science application.

3. Technology Stack

The platform is built using a combination of data science, machine learning, and visualization tools.

Category	Technology
Programming Language	Python
Web Framework	Streamlit
Data Analysis	Pandas
Visualization	Plotly, Matplotlib, Seaborn
Machine Learning	Scikit-learn
Report Generation	ReportLab
Business Intelligence	Power BI

These technologies together create a scalable and interactive analytics environment.

4. System Architecture

The system architecture follows a three-layer analytical architecture.

1. Data Layer

The data layer stores and manages the dataset containing information related to global income inequality indicators.

2. Analytics Layer

This layer performs statistical analysis, machine learning modeling, and automated insights generation.

3. Presentation Layer

The presentation layer provides an interactive dashboard interface developed using Streamlit.

Users interact with the platform through multiple modules including dashboards, visualization panels, and predictive models.

5. Dataset Description

The dataset used in the project contains economic indicators related to global income distribution.

Key attributes include

- Country Name
- Year
- Population indicators
- Income distribution metrics
- Economic inequality indicators

The dataset enables analysis of income inequality trends across countries and time periods.

6. Authentication System

The platform includes a secure login system implemented using password hashing techniques.

Features

- user authentication
- encrypted password verification
- session security
- restricted dashboard access

This ensures that only authorized users can access the analytics system.

7. Executive Dashboard

The Executive Dashboard provides a high-level overview of the dataset.

Key Metrics Displayed

- number of dataset rows
- number of dataset columns
- number of numerical variables
- number of categorical variables
- dataset quality score

The dashboard also includes graphical insights such as:

- data distribution charts
- categorical variable analysis
- correlation heatmaps

This module helps users quickly understand the structure of the dataset.

8. Power BI Dashboard Integration

The system integrates a business intelligence dashboard created in Power BI. The Power BI report is embedded directly inside the Streamlit application using an iframe

Benefits

- enterprise-level visualization
- interactive filtering

- real-time dashboard display

This integration combines BI reporting with data science capabilities.

9. Dataset Explorer

The Dataset Explorer module enables deep exploration of the dataset.

Features

- dataset overview
- missing value detection
- duplicate row detection
- skewness and kurtosis analysis
- outlier detection using IQR method
- correlation heatmap generation

The module also generates AI-based insights and data cleaning recommendations.

10. Advanced Visualization System

The platform includes a flexible visualization system that allows users to generate multiple chart types.

Supported charts include:

- histogram
- scatter plot
- line chart
- bar chart
- box plot
- violin plot
- bubble chart
- treemap
- sunburst chart
- density contour
- correlation heatmap
- 3D scatter plots

Users can customize chart appearance using color themes and chart templates.

11. Country Comparison Dashboard

The Country Comparison module allows users to analyze multiple countries simultaneously.

Users can select:

- multiple countries
- economic indicators

The module generates several comparative visualizations such as:

- grouped bar charts
- trend line graphs
- radar charts
- bubble relationship charts
- distribution box plots
- correlation heatmaps

This helps identify patterns and differences between countries.

12. Country Analysis Module

This module provides a detailed analysis for a single selected country.

Features include:

- country-specific dataset view
- historical trend analysis
- income distribution charts
- indicator relationship analysis
- statistical summaries

It enables deeper insights into individual country performance.

13. Global Map Visualization

The platform includes geographic visualizations to display inequality indicators worldwide.

Two types of maps are used:

3D Globe Map

Displays global data points on a three-dimensional earth model.

Choropleth Map

Uses color intensity to represent inequality indicators across countries.

These visualizations help users understand global patterns of income distribution.

14. AI Insights Generator

The AI Insights module automatically analyzes numerical variables and generates statistical summaries.

It calculates:

- mean
- median
- standard deviation
- minimum and maximum values

The system also detects:

- skewed distributions
- correlations
- outliers

Finally, it generates textual insights describing the behavior of the dataset.

14. Machine Learning Prediction Module

The platform integrates supervised machine learning models for predicting economic indicators.

Implemented models include:

- Linear Regression
- Decision Tree Regression
- Random Forest Regression
- Gradient Boosting Regression

The process includes:

- selecting target variable
- training models
- evaluating performance using R^2 score
- selecting the best model

The module also visualizes:

- model comparison
- feature importance
- actual vs predicted values
- residual errors

15. AutoML System

The AutoML module automatically evaluates multiple machine learning models and selects the best one.

Algorithms evaluated include:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- K-Nearest Neighbors

Each model is evaluated using:

- R^2 Score
- Mean Absolute Error
- Root Mean Square Error

The system automatically selects the highest performing model.

16. Time Series Forecasting

The forecasting module predicts future values of selected indicators.

Key capabilities include:

- time series visualization
- moving average trend analysis
- regression-based forecasting
- future value prediction

The system also determines whether the trend is:

- increasing
- decreasing
- stable

17. KPI Alert System

The KPI alert system detects anomalies in dataset indicators.

It uses anomaly detection techniques such as Isolation Forest to identify unusual values and notify users.

18. Automated Report Generation

The platform allows users to export analytical summaries as PDF reports.

Reports include:

- dataset statistics
- analytical summaries
- key insights

Reports are generated dynamically using Python libraries.

19. FAQ Section

The FAQ page explains common questions about the platform including:

- purpose of the system
- dataset information
- explanation of inequality indicators
- technology stack used

This improves usability for new users.

20. About Page

The About page provides an overview of the Global Income Intelligence Platform including:

- project objectives
- platform features
- technology used
- future development possibilities

21. Deployment

The Streamlit application is deployed as a web-based analytics platform.

Deployment Steps:

- Prepare project files
- Install dependencies
- Configure Streamlit application
- Deploy on hosting platform
- Test application functionality

Once deployed, the system becomes accessible through a public web link.

22. Benefits of the Platform

The Global Income Intelligence Platform provides several advantages:

- interactive data exploration
- automated analytics insights
- machine learning integration
- real-time visualization
- predictive analytics capabilities

The platform combines the strengths of business intelligence and data science.

23. Project Outcomes

The final system successfully integrates:

- advanced data visualization
- machine learning prediction models
- automated data analysis
- geographic visualizations
- forecasting tools
- interactive dashboards

The platform provides a comprehensive analytical environment for global income inequality research.

23. Conclusion

Milestone 3 successfully transformed the project from a traditional dashboard into a fully interactive analytics platform.

By combining Power BI dashboards with a Streamlit web application, the project demonstrates how modern data science technologies can be integrated to build powerful analytical tools.

The Global Income Intelligence Platform enables users to explore global inequality data, generate insights, and make predictions using machine learning techniques.

This milestone highlights the potential of combining business intelligence, interactive visualization, and predictive analytics within a single scalable system.

24. System Workflow Diagram

The Global Income Intelligence Platform follows a structured workflow that processes data from ingestion to visualization and predictive analytics.

Workflow Steps

Step 1 – Data Collection

The system imports the dataset containing global income inequality indicators. The dataset includes economic indicators, population metrics, and country-level statistics.

Step 2 – Data Processing

The dataset is cleaned and processed using Pandas. This stage includes:
handling missing values
detecting duplicates
statistical summary generation
feature extraction

Step 3 – Exploratory Data Analysis

Exploratory analysis is performed to understand dataset patterns using visualization tools such as Plotly and Matplotlib.

The analysis includes:

- distribution analysis
- correlation analysis
- outlier detection
- feature importance evaluation

Step 4 – Visualization Layer

Interactive dashboards are created using Streamlit. Users can generate dynamic charts, maps, and comparisons across countries.

Step 5 – Machine Learning Processing

Predictive models are trained using Scikit-learn algorithms. These models learn patterns in the dataset and predict future values.

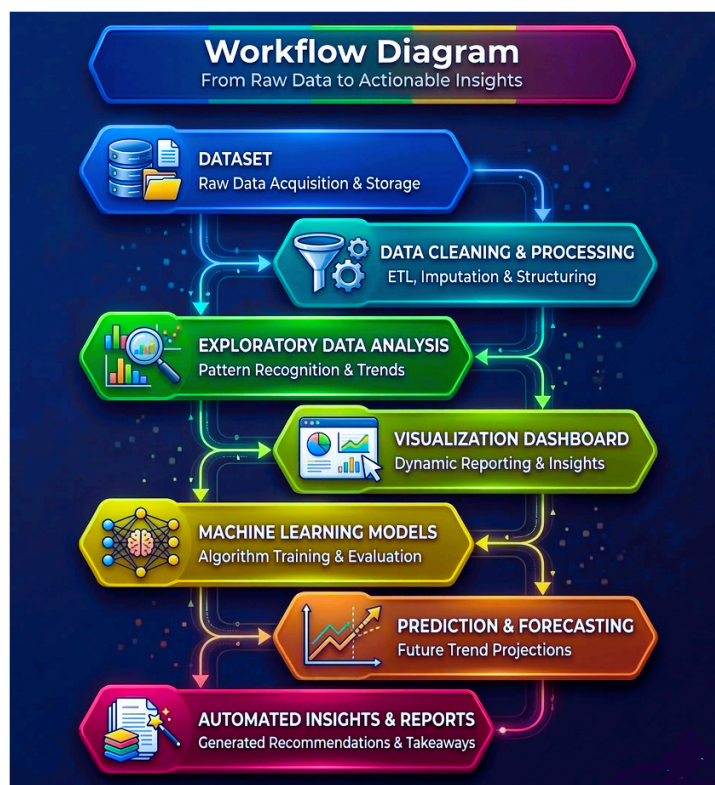
Step 6 – Forecasting and Prediction

Time series forecasting is applied to estimate future trends in income inequality indicators.

Step 7 – Insights and Reporting

The system generates automatic insights and exports analytical reports in PDF format.

Workflow Diagram



25. Machine Learning Algorithms Used

The platform integrates several machine learning algorithms to perform predictive analytics on income inequality indicators.

All models are implemented using Scikit-learn.

1. Linear Regression

Linear Regression is a statistical model used to predict continuous numerical values.

Formula

$$Y = a + bX$$

Where:

Y = predicted value

X = input feature

a = intercept

b = slope

Purpose in the Project

Linear regression predicts economic indicators such as income inequality values based on other variables in the dataset.

2. Decision Tree Regression

Decision Tree Regression divides the dataset into smaller subsets based on decision rules.

Key Characteristics:

1. tree-based model
2. handles nonlinear relationships
3. interpretable structure

Usage in the Platform:

It predicts inequality indicators by learning patterns from historical data.

3. Random Forest Regression

Random Forest is an ensemble learning method that combines multiple decision trees.

Advantages:

high prediction accuracy
reduced overfitting
robust against noise

Role in the Project:

The model analyzes multiple dataset features simultaneously to generate more reliable predictions.

4. Gradient Boosting Regression

Gradient Boosting improves model performance by sequentially correcting prediction errors.

Features:

strong predictive power
optimized learning process
high accuracy

Application:

It refines predictions for economic indicators by minimizing model errors during training.

5. K-Nearest Neighbors (KNN)

KNN predicts values based on similarity between data points.

Working Principle:

The algorithm finds the nearest data points and calculates predictions based on their values.

Application:

KNN helps in predicting values based on similar country profiles.

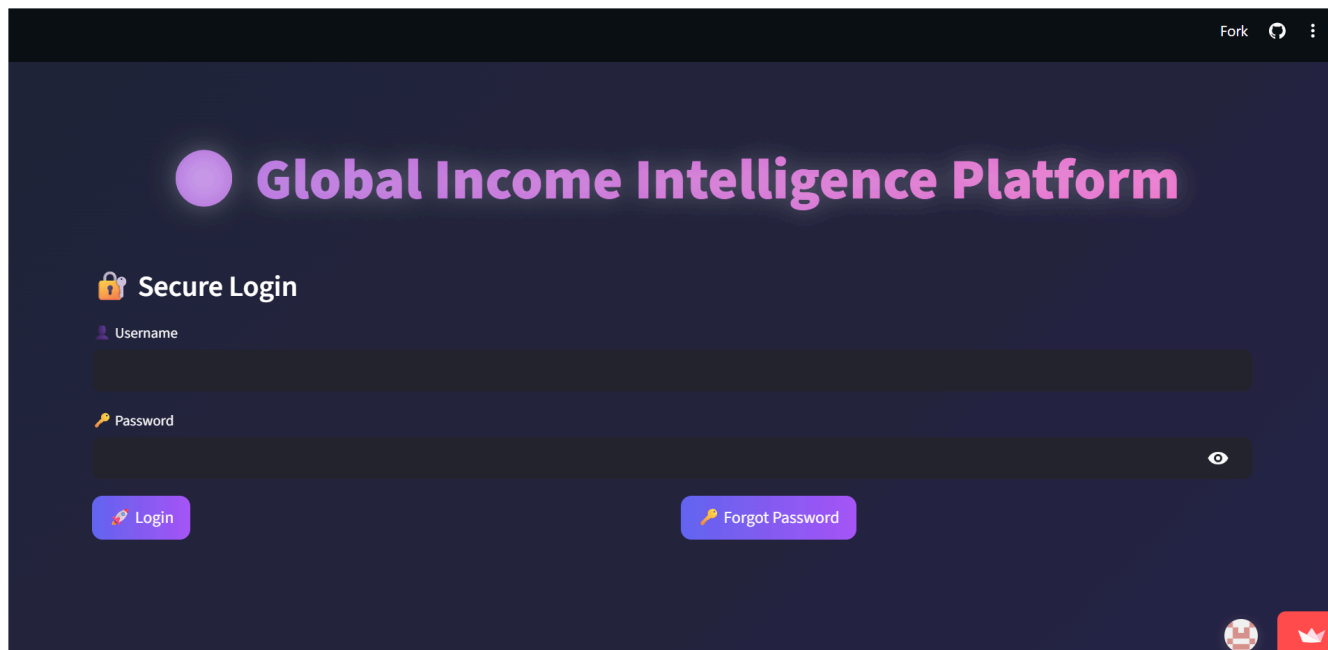
26. Screenshots Explanation of Dashboard Modules

1. Login Page

The login page provides secure authentication for accessing the platform.

Features:

password verification
session security
protected dashboard access

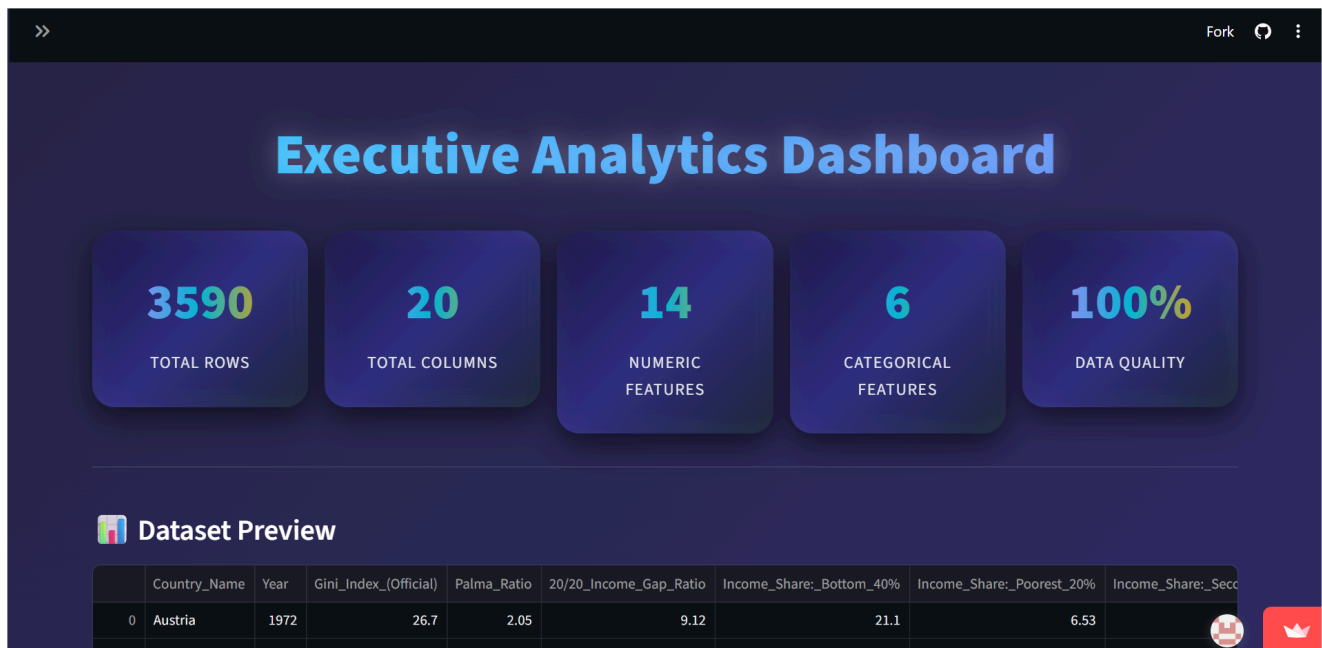


2. Executive Dashboard

The Executive Dashboard provides an overview of the dataset.

Displays:
dataset statistics
distribution charts
correlation heatmap
categorical analysis

This dashboard helps users quickly understand the structure of the dataset.

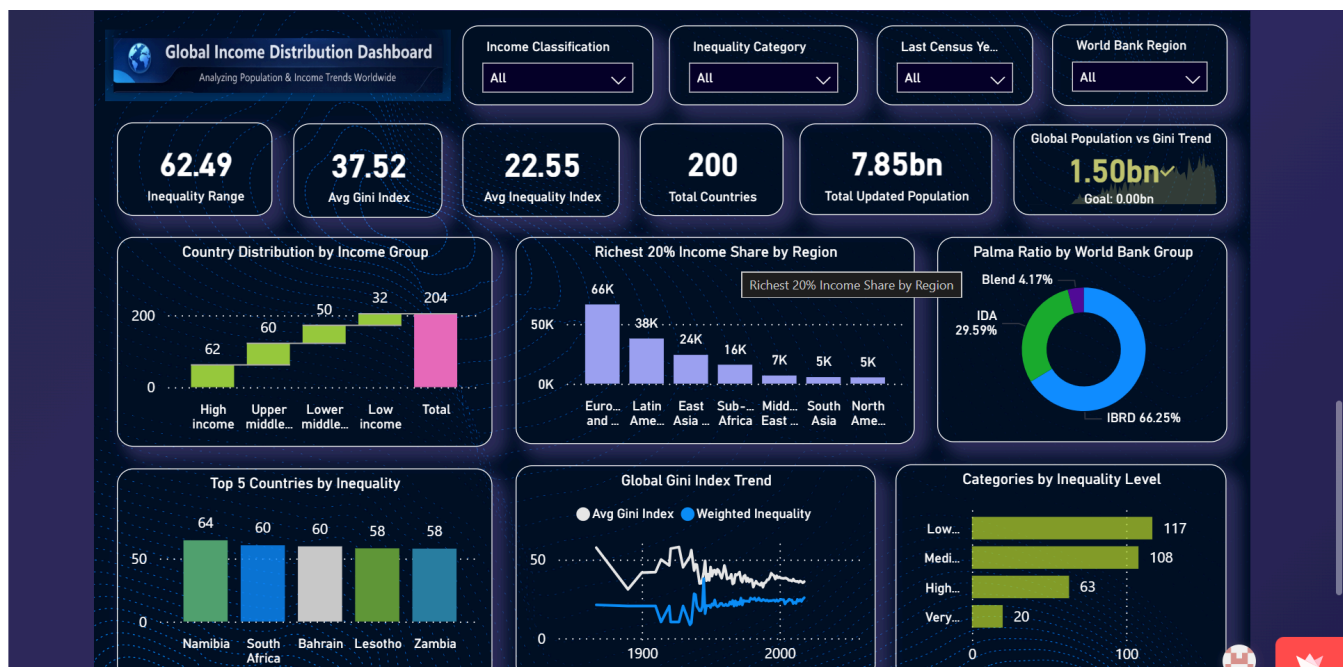


3. Power BI Dashboard Integration

The platform embeds a dashboard created in Microsoft Power BI.

Features:
interactive filters
enterprise visualization
embedded analytics

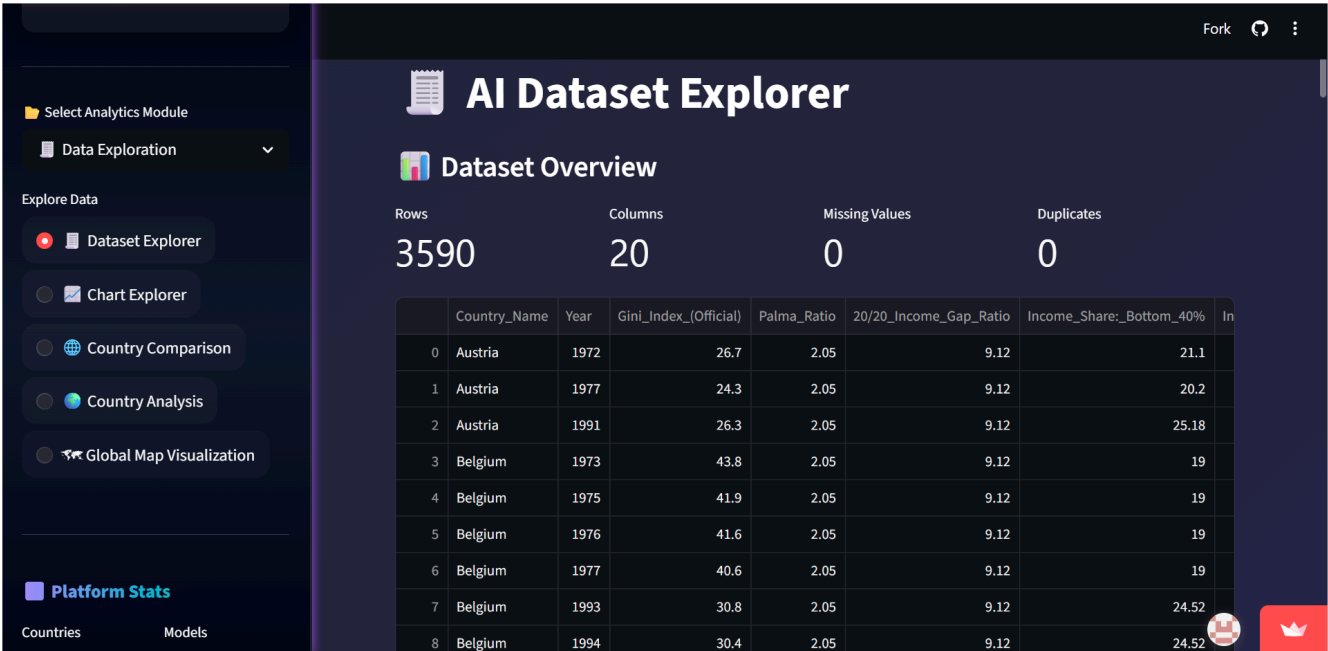
This integration combines business intelligence with Python-based analytics.



4. Dataset Explorer

The Dataset Explorer allows users to inspect the dataset.

- Functions:
- view dataset records
 - detect missing values
 - identify duplicates
 - analyze statistical properties

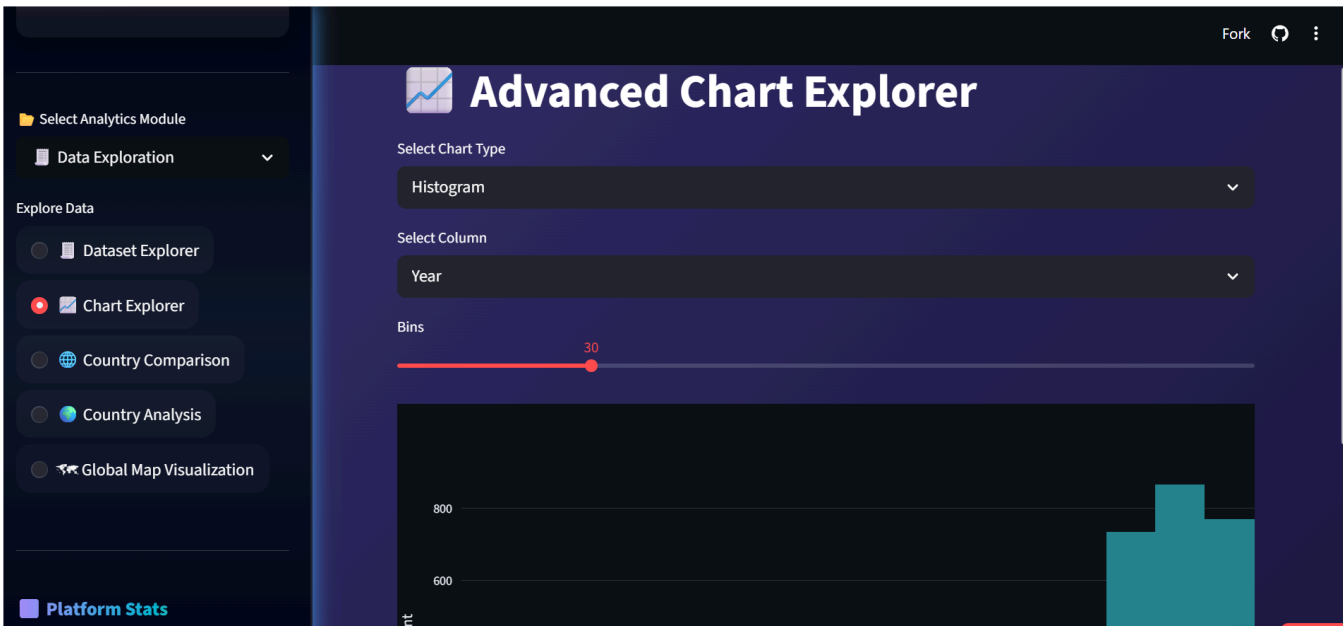


5. Visualization Explorer

This module allows users to generate interactive charts.

- Supported 15+ Charts:
- scatter plots
 - histograms
 - line charts
 - box plots

Users can customize charts dynamically.



6. Country Comparison Dashboard

This module allows users to compare multiple countries.

Visualizations:
radar charts
grouped bar charts
trend lines
correlation heatmaps

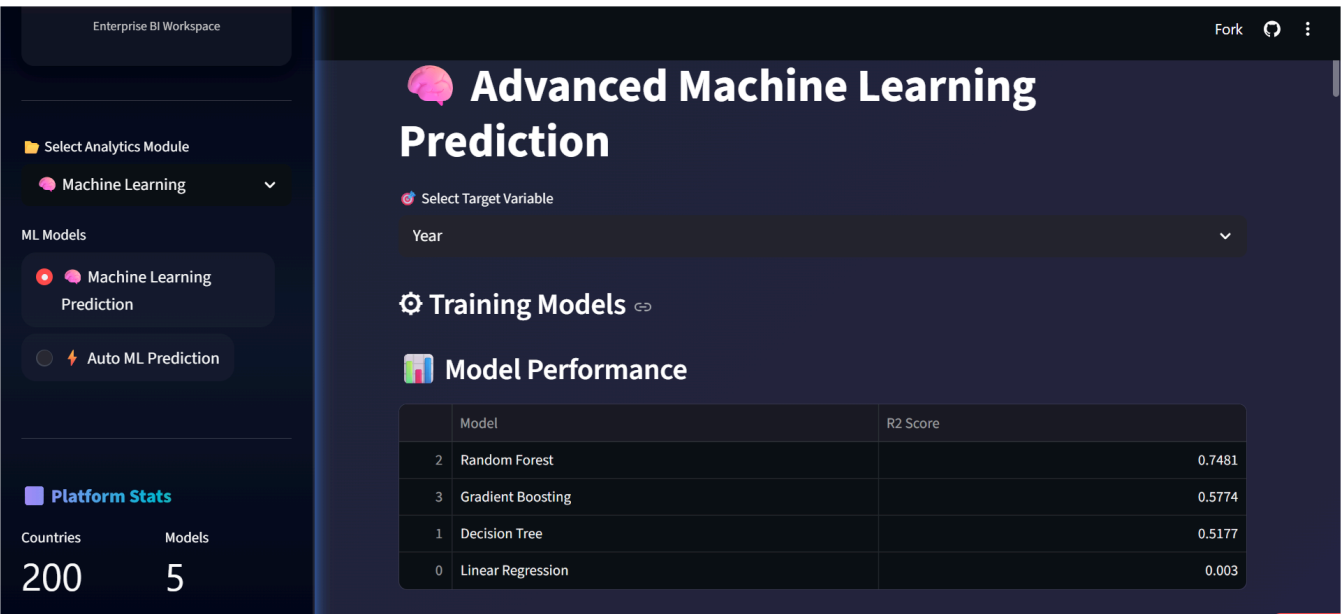
It highlights differences between economic indicators across countries.



7. Machine Learning Prediction Page

This page allows users to train models and generate predictions.

Features:
model selection
prediction visualization
feature importance charts
actual vs predicted comparison



8. Forecasting Dashboard

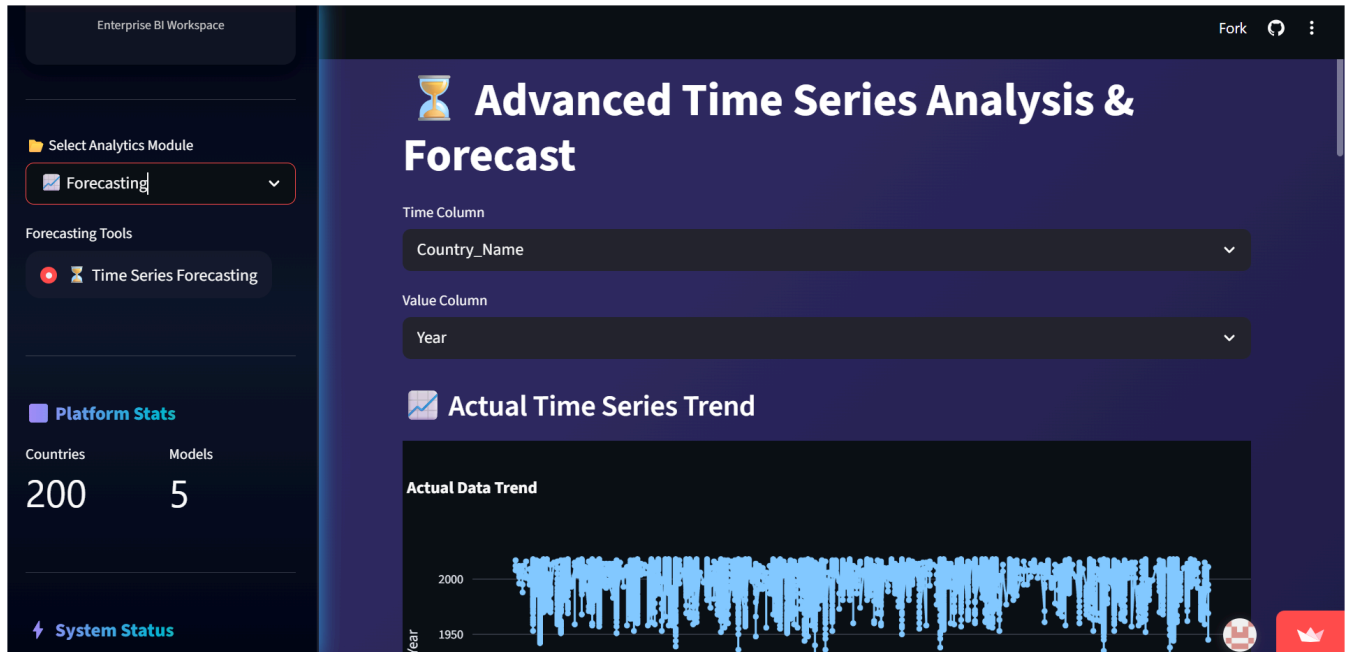
The forecasting module predicts future trends.

Visualizations:

historical trend analysis

moving averages

predicted future values



27. Final link of Power BI Dashboard Embed app link

[https://app.powerbi.com/view?
r=eyJrIjojNGZlMTUzYTktODU3OC00ODgxLWE3ZmItZjlmM2Y2MTg5ZWQxliwidCI6IjNmMGQxMTRlTVmZjltNDk0NS04OThjLWRkZTk3Y2Y2NWZkNSJ9](https://app.powerbi.com/view?r=eyJrIjojNGZlMTUzYTktODU3OC00ODgxLWE3ZmItZjlmM2Y2MTg5ZWQxliwidCI6IjNmMGQxMTRlTVmZjltNDk0NS04OThjLWRkZTk3Y2Y2NWZkNSJ9)

28. Final link of streamlit app link

<https://global-income-distribution-project-4nyywmtrs38ed95pbcvhc4.streamlit.app>